This is my personal style guide/notation guide as of 2026. I am writing this primarily for my own benefit so I know what notation I will use as I comb through sources.

- Sets are denoted by standard Roman characters, such as $A, B, C$, or capitalized Greek characters such as $\Omega, \Lambda, \Gamma$.
- Vector spaces are denoted by `\mathscr`, such as $\mathscr{H}$ or $\mathscr{X}$.
- Nets or sequences are denoted by $(x_i)_{i\in I}$ or $(x_n)_{n\in\mathbb{N}}$.
- Indexed sets are written $\{x_i : i \in I\}$ or $\{x_n : n \in \mathbb{N}\}$.
- Families of subsets (such as topologies or otherwise) are denoted by `\mathcal`, such as $\mathcal{T}$ or $\mathcal{U}$.
- The system of open neighborhoods of a point $y \in X$ is denoted $\mathcal{O}_X(y)$. The system of all neighborhoods of a point $y \in X$ is denoted $\mathcal{N}_X(y)$.
- All vector spaces are complex unless stated otherwise.
- If $(X, d)$ is a metric space, then

$$\begin{aligned} U(x,\varepsilon) &= \{y \in X : d(x,y) < \varepsilon\} \\ B(x,\varepsilon) &= \{y \in X : d(x,y) \leq \varepsilon\} \\ S(x,\varepsilon) &= \{y \in X : d(x,y) = \varepsilon\}. \end{aligned}$$

- The Roman lettering $L^p$ spaces are used to denote almost-everywhere equivalence classes of $p$-integrable functions, while the `\mathcal` lettering $\mathcal{L}^p$ spaces are used to denote the space of all the $p$-integrable functions without considering almost-everywhere equivalence.
- If $\mathscr{X}$ is a normed vector space, then ball($\mathscr{X}$) is used to denote the *closed* unit ball of $\mathscr{X}$ about the origin.
- If $\mathscr{X}$ is a normed vector space, then $\mathcal{B}(\mathscr{X})$ denotes the space of bounded linear operators from $\mathscr{X}$ to itself.
- If $\mathscr{H}$ is a Hilbert space, then $\mathcal{U}(\mathscr{H})$ is the space of unitary operators on $\mathscr{H}$.
- Operators in $\mathcal{B}(\mathscr{H})$ for some Hilbert space $\mathscr{H}$ are denoted by standard Roman characters, such as $T, S, U$.
- If $\mathscr{K} \subseteq \mathscr{H}$ is a closed subspace, then proj($\mathscr{K}$) is used to denote the projection onto the subspace $\mathscr{K}$.
- If $p$ and $q$ are two projections, then $p \vee q$ is their supremum and $p \wedge q$ is their infimum. If $p$ and $q$ are orthogonal to each other, then we write $p \vee q = p + q$.
- $C^*$-Algebras or von Neumann algebras will be written with the `\mathcal` lettering, such as $\mathcal{A}$ or $\mathcal{M}$.
- If $A \subseteq \mathcal{B}(\mathscr{H})$ is any subset of operators, then $W^*(A)$ is used to denote the von Neumann algebra generated by the set $A$.
- If $\mathcal{M}$ is a von Neumann algebra, then Proj($\mathcal{M}$) is used for the projection lattice of $\mathcal{M}$.